

Multiple Choice

1. **Answer:** A

Options: A) Smaller cells are more resilient to external damage.; B) Smaller cells have a larger number of organelles.; C) Smaller cells require less energy to function.; D) Smaller cells can multiply faster.; E) Smaller cells fit together more tightly.

Note: Correct option: A - Smaller cells are more resilient to external damage.

2. **Answer:** E

Options: A) disruptive selection; B) sexual selection; C) bottleneck effect; D) non-random mating; E) stabilizing selection

Note: Correct option: E - stabilizing selection

3. **Answer:** A

Options: A) Through vertical gene transfer from parent to offspring; B) Through random assortment of genes during meiosis; C) Through peptide bond formation errors during translation; D) Through genetic mutation only; E) Through transcriptional errors in RNA processing

Note: Correct option: A - Through vertical gene transfer from parent to offspring

4. **Answer:** E

Options: A) By using a controlled group of dogs not exposed to the sounds; B) By using sensory deprivation to heighten hearing sensitivity; C) By observing the dog's natural behavior in different sound environments; D) By using negative reinforcement only; E) By using operant conditioning

Note: Correct option: E - By using operant conditioning

5. **Answer:** A

Options: A) -.27, 1.45, 1.73; B) -.27, -1.45, -1.73; C) -2.18, 0.15, -1.82; D) 0, 1.5, -2; E) .27, 1.45, 1.73

Note: Correct option: A - -.27, 1.45, 1.73

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Chemical shift subtraction can contribute to routine dynamic contrast subtraction in morphologic analysis particularly for the evaluation of margins of benign lesions in fatty breasts. It can also help in morphologic analysis of masses in dense breast.

7. **Answer:** True

Options: A) True; B) False

Note: It can be concluded that CLBP patients in general have worse motor task performance compared to healthy subjects and that provoking pain-related cognitions further worsened performance.

8. **Answer:** True

Options: A) True; B) False

Note: For a given NIHSS score, the median volume of right hemisphere strokes is consistently larger than the median volume of left hemisphere strokes. The clinical implications of our finding need further exploration.

9. **Answer:** False

Options: A) True; B) False

Note: Consistent with prior studies, we found an inverse relationship between obesity and serum prostate specific antigen. However, the magnitude of the difference was small. Thus, adjusting prostate specific antigen for body mass index does not appear warranted.

10. **Answer:** False

Options: A) True; B) False

Note: Our results suggest that colonoscopy does not affect the management of patients with acute diverticulitis nor alter the outcome. The current practice of a routine colonoscopy after acute diverticulitis, diagnosed by typical clinical symptoms and CT needs to be reevaluated.

Long Form Response

11. **Answer:** No, climatologists would find it more helpful to correspond with a geologist.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** The neurotransmitter is present in the axonal end of a neuron only and there are no synaptic vesicles inside the dendrites of a neuron.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key